

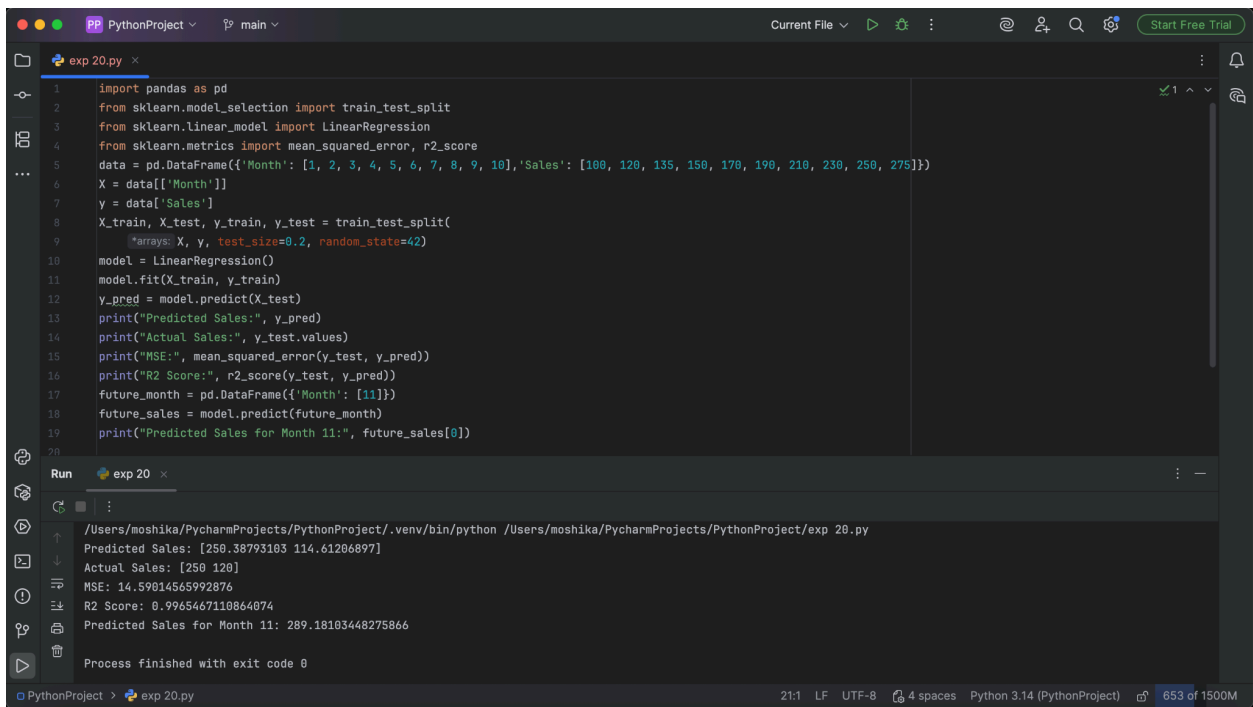
EXPERIMENT - 20

Aim: To predict future sales using a suitable machine learning regression algorithm.

Procedure:

1. Load the sales dataset.
2. Clean and preprocess the data.
3. Extract relevant features such as date, product, and previous sales.
4. Separate input features and sales values.
5. Split the data into training and testing sets.
6. Train a suitable regression model.
7. Predict future sales.
8. Evaluate the prediction performance.
9. Display the predicted sales values.

Output:



```
1 import pandas as pd
2 from sklearn.model_selection import train_test_split
3 from sklearn.linear_model import LinearRegression
4 from sklearn.metrics import mean_squared_error, r2_score
5 data = pd.DataFrame({'Month': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10], 'Sales': [100, 120, 135, 150, 170, 190, 210, 230, 250, 275]})
6 X = data[['Month']]
7 y = data['Sales']
8 X_train, X_test, y_train, y_test = train_test_split(
9     *arrays: X, y, test_size=0.2, random_state=42)
10 model = LinearRegression()
11 model.fit(X_train, y_train)
12 y_pred = model.predict(X_test)
13 print("Predicted Sales:", y_pred)
14 print("Actual Sales:", y_test.values)
15 print("MSE:", mean_squared_error(y_test, y_pred))
16 print("R2 Score:", r2_score(y_test, y_pred))
17 future_month = pd.DataFrame({'Month': [11]})
18 future_sales = model.predict(future_month)
19 print("Predicted Sales for Month 11:", future_sales[0])
20
```

Run exp 20

```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/exp_20.py
Predicted Sales: [250.38793183 114.6126897]
Actual Sales: [250 120]
MSE: 14.59814565992876
R2 Score: 0.9965467110864074
Predicted Sales for Month 11: 289.18103448275866
Process finished with exit code 0
```

Result: The machine learning model successfully predicted future sales based on historical sales data.